Mixed Strategies and Nash Equilibria

2026-04-13

Beyond Pure Strategies

Plan

- So far, strategies have been **pure**: choose one action deterministically.
- Today we allow players to **randomise** by playing a **mixed strategy**.
- This requires cardinal payoffs, connects back to expected utility, and completes the Nash equilibrium story.

Associated Reading

Game Theory, Sections 6.1–6.3

Why Randomise?

In some games, any deterministic choice can be exploited.

Consider **Matching Pennies**:

	Heads	Tails
Heads	(+1, −1)	(−1, +1)
Tails	(−1, +1)	(+1, −1)

If Row plays Heads for certain, Column plays Tails. If Row plays Tails for certain, Column plays Heads. There is **no pure strategy Nash equilibrium**.

If Row plays Heads with probability p and Tails with probability $1 - p$:

Column's expected payoff from Heads: $-p + (1 - p) = 1 - 2p$

Column's expected payoff from Tails: $p - (1 - p) = 2p - 1$

Column is indifferent when $1 - 2p = 2p - 1$, i.e., $p = 1/2$.

At $p = 1/2$, Column cannot exploit Row's strategy. Any mix gives the same expected payoff. The **mixed strategy Nash equilibrium** is for both players to mix 50/50.

Mixed Strategy Nash Equilibrium

Definition

A **mixed strategy** assigns a probability to each pure strategy.

A **mixed strategy Nash equilibrium** is a profile of (possibly mixed) strategies such that no player can improve their *expected* payoff by unilaterally deviating.

Nash's Theorem (1950): Every finite game has at least one Nash equilibrium in pure or mixed strategies.

In a mixed strategy Nash equilibrium, a player who mixes must be **indifferent** between the strategies they mix over.

Why? If one strategy had higher expected payoff than another, the player would shift all probability to that strategy (pure strategy, not a mix).

This gives us a way to find mixed equilibria: **set the opponent indifferent**.

	Opera	Football
Opera	(2, 1)	(0, 0)
Football	(0, 0)	(1, 2)

We found two pure strategy Nash equilibria: (Opera, Opera) and (Football, Football).

Is there also a mixed equilibrium?

Let Row play Opera with probability p.

Column is indifferent when:

$$1 \cdot p + 0 \cdot (1 - p) = 0 \cdot p + 2 \cdot (1 - p)$$
$$p = 2 - 2p \implies p = \tfrac{2}{3}$$

Let Column play Opera with probability q.

Row is indifferent when:

$$2q + 0(1 - q) = 0 \cdot q + 1 \cdot (1 - q)$$
$$2q = 1 - q \implies q = \tfrac{1}{3}$$

Mixed equilibrium: Row plays Opera with probability $\frac{2}{3}$; Column plays

Row's expected payoff:

$$2 \cdot \frac{1}{3} + 0 \cdot \frac{2}{3} = \frac{2}{3}$$

Column's expected payoff:

$$1 \cdot \frac{2}{3} + 0 \cdot \frac{1}{3} = \frac{2}{3}$$

Both do **worse** in the mixed equilibrium ($\frac{2}{3}$) than in either pure equilibrium (Row gets 2 or 1; Column gets 1 or 2).

The mixed equilibrium is the outcome when players can't coordinate on which pure equilibrium to play.

Cardinal Payoffs and Zero-Sum Games

For pure strategy analysis, ordinal payoffs suffice. We just need to know which outcomes are ranked higher.

For mixed strategies, we need **cardinal** payoffs. The actual numbers matter, not just the order.

This is because mixed strategy analysis is expected utility calculation. The magnitudes of payoffs affect the optimal mixing probabilities.

A **zero-sum game** is one where every gain for one player is an equal loss for another:

$$u_1(s) + u_2(s) = 0 \text{ for all strategy profiles } s$$

Matching Pennies is zero-sum. Poker is (approximately) zero-sum.

In zero-sum games, the interests of players are **perfectly opposed**.

The Minimax Theorem

For two-player zero-sum games:

> ***von Neumann's Minimax Theorem*** *(1928): There is always a mixed strategy profile where each player* ***maximises their minimum guaranteed payoff*** *(the minimax strategy).*

The minimax strategy and the Nash equilibrium coincide in zero-sum games.

This was the first major theorem of game theory, proved before Nash's more general result.

Each player's minimax strategy: mix 50/50. This guarantees an expected payoff of 0.

Against any deterministic opponent, you can't do better on average, but you can't do worse either.

In zero-sum games, the minimax strategy is the uniquely rational strategy.

Interpreting Mixed Strategies

What Does Mixing Mean?

Players don’t literally flip coins at the start of a game (usually).

Several interpretations:

- **Deliberate randomisation**: a player genuinely randomises to stay unpredictable (relevant in sports, military strategy).
- **Population interpretation**: the equilibrium describes a *distribution* across a large population of players, not an individual’s behaviour.

Mixed Strategies in Practice

Mixed strategy equilibria are especially relevant when:

- **Unpredictability has strategic value**: penalty kicks in soccer, play-calling in American football, auditing and compliance.
- **Coordination fails**: the mixed equilibrium is the “default” when no focal pure equilibrium exists.

Worked Example

Two firms simultaneously choose whether to adopt **Standard A** or **Standard B** for a new product. Each prefers that the other adopts the same standard (compatibility is valuable), but Firm 1 has a slight preference for A and Firm 2 for B.

	Firm 2: A	Firm 2: B
Firm 1: A	(3, 2)	(0, 0)
Firm 1: B	(0, 0)	(2, 3)

Step 1: Find Pure-Strategy NE

(A, A): Firm 1 gets 3, switch to B gives 0. No improvement. Firm 2 gets 2, switch to B gives 0. No improvement. **NE.**

(B, B): Firm 1 gets 2, switch to A gives 0. No improvement. Firm 2 gets 3, switch to A gives 0. No improvement. **NE.**

(A, B) and **(B, A)**: both give (0, 0), and both firms can improve by switching. **Not NE.**

Two pure equilibria. Is there also a mixed one?

Let Firm 1 play A with probability p and Firm 2 play A with probability q.

Make Firm 1 indifferent (Firm 1's payoff from A = payoff from B):

$$3q + 0(1-q) = 0 \cdot q + 2(1-q)$$

$$3q = 2 - 2q$$

$$5q = 2 \implies q = \frac{2}{5}$$

Make Firm 2 indifferent (Firm 2's payoff from A = payoff from B):

$$2p + 0(1 - p) = 0 \cdot p + 3(1 - p)$$

$$2p = 3 - 3p$$

$$5p = 3 \implies p = \frac{3}{5}$$

Mixed equilibrium: Firm 1 plays A with probability $\frac{3}{5}$; Firm 2 plays A with probability $\frac{2}{5}$.

Step 3: Expected Payoffs

Firm 1's expected payoff (using either pure strategy, since indifferent):

$$3 \cdot \frac{2}{5} + 0 \cdot \frac{3}{5} = \frac{6}{5} = 1.2$$

Firm 2's expected payoff:

$$2 \cdot \frac{3}{5} + 0 \cdot \frac{2}{5} = \frac{6}{5} = 1.2$$

Both get 1.2 in the mixed equilibrium, worse than *either* pure equilibrium (where someone gets 2 and someone gets 3).

The mixed equilibrium represents the outcome when coordination fails. Each firm leans toward its preferred standard, and sometimes they end up mismatched.

Worked Example: Penalty Kicks

Penalty Kicks as a Zero-Sum Game

Research on penalty kicks in soccer confirms that kickers and goalies mix their strategies in ways that match equilibrium predictions (See, for instance, Palacios-Huerta 2003, “Professionals Play Minimax”, Review of Economic Studies).

We can model a penalty kick as a zero-sum game. The kicker and the goalie each choose Left, Centre, or Right (directions from the goalie’s perspective). Choices are simultaneous: the goalie commits before seeing where the ball goes.

For a right-footed kicker:

- Kicking across the body (to goalie's right) generates more power and accuracy than kicking to the natural side (goalie's left).
- When the goalie guesses the correct side, the save is more likely, but across-body kicks are still harder to stop.
- A kick down the middle is saved with certainty if the goalie stays put.

Payoffs are the kicker's scoring probability. The goalie's payoff is the negative of each entry, since this is a zero-sum game.

	Goalie L	Goalie C	Goalie R
Kick L	.4	.8	.8
Kick C	.8	0	.8
Kick R	.9	.9	.5

Kick R (across body) scores at a higher rate than Kick L (natural side) whether goalie guesses correctly or incorrectly. But the kicker can't just always kick right, or the goalie adjusts.

No Pure Strategy Equilibrium

Whatever the kicker picks, the goalie can improve by adjusting. Whatever the goalie picks, the kicker can improve by adjusting.

For example: if the kicker always goes right, the goalie goes right and holds scoring to .5. Then the kicker switches to left, scoring .8. Then the goalie follows, and so on.

We need a mixed strategy equilibrium.

Finding the Goalie's Equilibrium Mix

Let the goalie play L, C, R with probabilities q, s, and $1 - q - s$.

The indifference principle says the kicker must get the same expected payoff from each option. (We won't expect you to solve these kinds of equations on the test, but it's the kind of thing you can do.)

Kick L: $0.4q + 0.8s + 0.8(1 - q - s) = 0.8 - 0.4q$

Kick C: $0.8q + 0 \cdot s + 0.8(1 - q - s) = 0.8 - 0.8s$

Kick R: $0.9q + 0.9s + 0.5(1 - q - s) = 0.5 + 0.4q + 0.4s$

Solving for the Goalie's Mix

Setting Kick L = Kick C:

$$0.8 - 0.4q = 0.8 - 0.8s$$

$$0.4q = 0.8s \implies q = 2s$$

Setting Kick L = Kick R:

$$0.8 - 0.4q = 0.5 + 0.4q + 0.4s$$

$$0.3 = 0.8q + 0.4s$$

Substituting $q = 2s$: we get $0.3 = 2s$, so $s = 0.15$ and $q = 0.30$.

Goalie's equilibrium mix: L = 30%, C = 15%, R = 55%.

Finding the Kicker's Equilibrium Mix

Let the kicker play L, C, R with probabilities p, r, and $1 - p - r$.

Now the goalie must be indifferent. The expected scoring rate must be the same regardless of what the goalie does.

Setting the scoring rate against Goalie L equal to Goalie C:

$$0.4p + 0.8r + 0.9(1 - p - r) = 0.8p + 0 \cdot r + 0.9(1 - p - r)$$

$$0.8r = 0.4p \implies p = 2r$$

Solving for the Kicker's Mix

Setting the scoring rate against Goalie L equal to Goalie R:

$$0.4p + 0.8r + 0.9(1 - p - r) = 0.8p + 0.8r + 0.5(1 - p - r)$$

$$0.4 = 0.8p + 0.4r$$

Substituting $p = 2r$: we get $0.4 = 2r$, so $r = 0.2$ and $p = 0.4$.

Kicker's equilibrium mix: L = 40%, C = 20%, R = 40%.

The Equilibrium

Player	Left	Centre	Right
Kicker	40%	20%	40%
Goalie	30%	15%	55%

The equilibrium scoring rate is **68%**, regardless of what either player does (given the other plays the equilibrium mix). Actual penalties are scored at more like 80%, but I've picked these numbers to make the math easier. For real numbers, see the actual research papers!

Two things to notice about this. The kicker kicks left and right equally often, even though across-body kicks are better, because the goalie compensates by going right 55% of the time. And the kicker goes down the middle one time in five, even though it's a guaranteed save if the goalie reads it, because it keeps the goalie from committing to the sides.

For Next Time

- We’ll do some revision, and some more examples of mixed equilibria.